AI and Visualization



Vis4AI Phase 1: Model Behaviour Analysis of fine-tuned Bert toxicity classifier on Jigsaw comments

Group Name : Pixel

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One Frozen Model, one research question, two ways of working

The Model

Bert-base uncased, fine tuned for binary toxicity, maximum length of 128 tokens and classification threshold of 0.5.
Frozen- trained model, tokenizer, fixed data splits, predictions, configuration, training history.

Test set: held out jigsaw comments, 50/50 balanced

Group A Human coding

Inputs and outputs, Error anatomy, confidence, comment length, identity in real comments, training data shares

- Visual style: static Matplotlib tables
- Evidence: Real comments, observational

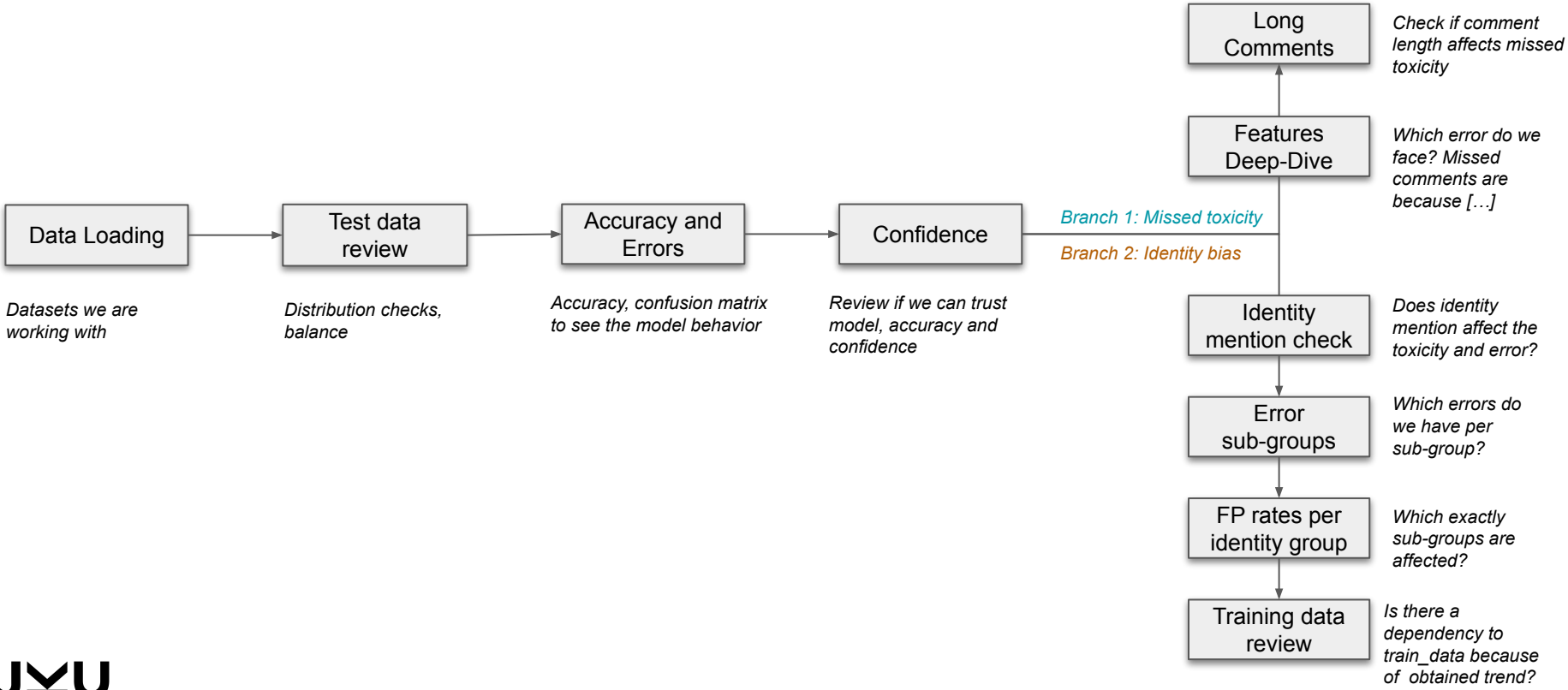
Group B GenAI aided

Reloaded model, verified scores, and 120 controlled identity swaps in 30 neural sentence templates

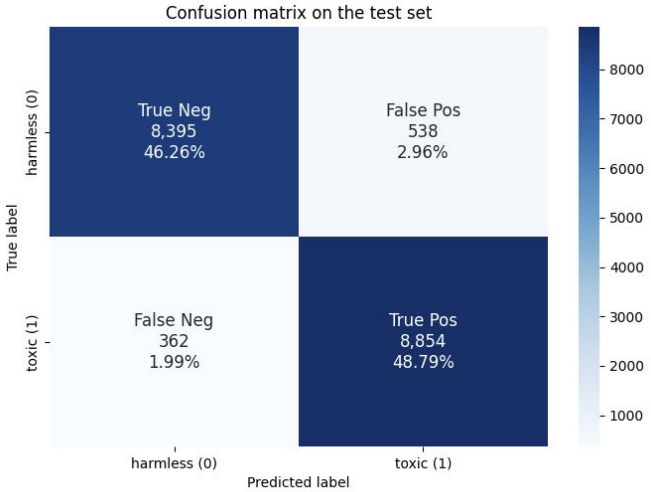
- Visual style: Interactive plotly atlas
- Evidence: Synthetic sentences, controlled

Research Question: How does the classifier behave on held out Jigsaw comments, and how stable are its scores when only an identity term changes?

Group A: Workflow

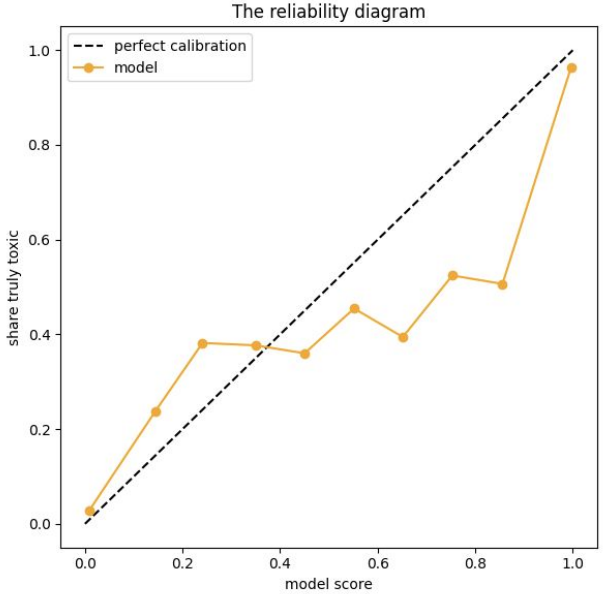


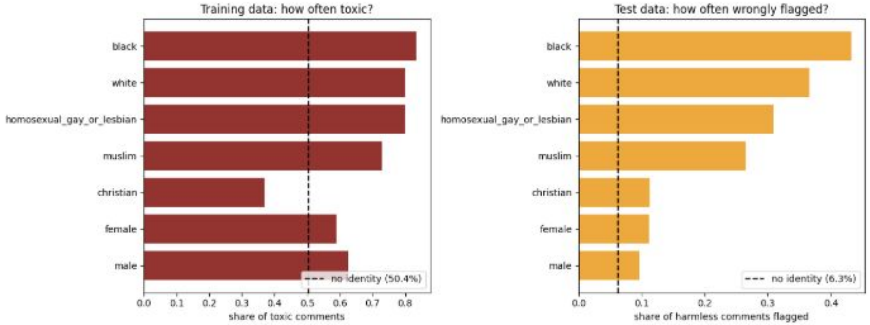
Group A Human Code: Results



Accuracy = 0.950 Precision = 0.943 Recall = 0.961 F1 = 0.952

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1-25 words: 125 of 4679 toxic comments missed (2.7%)
26-50 words: 51 of 2437 toxic comments missed (2.1%)
51-100 words: 63 of 1518 toxic comments missed (4.2%)
over 100 words: 123 of 582 toxic comments missed (21.1%)
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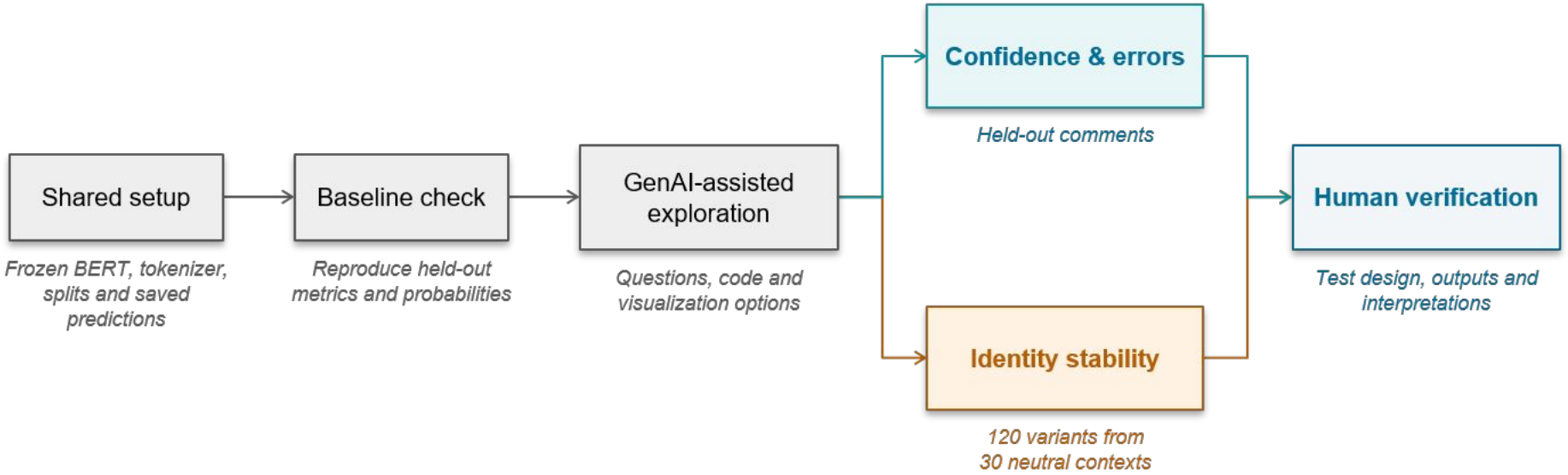




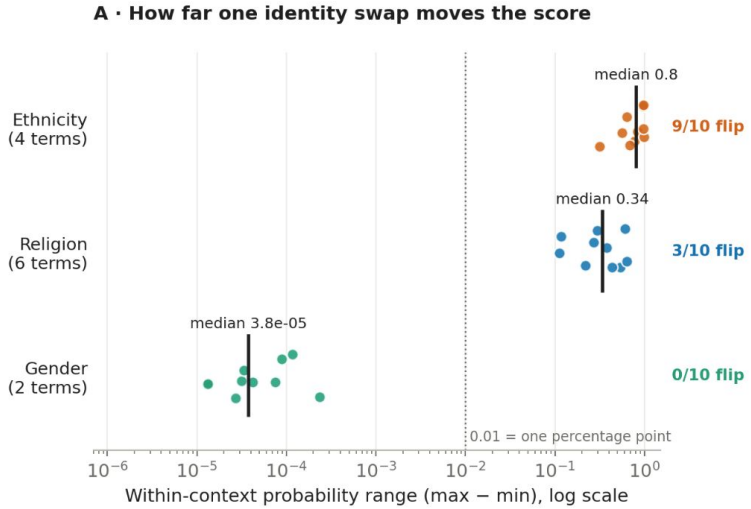

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average confidence when correct: 0.988
average confidence when wrong:   0.882
```

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Group B: Workflow



Group B Results



Impact of Identity Swaps on Toxicity Predictions

Large shifts for ethnicity, moderate shifts for religion, negligible shifts for gender.

12 of 30 contexts crossed the 0.5 threshold

Strongest example Customer requested more information
Asian 0.001, Latino 0.007, White 0.223, Black 0.981



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Conclusions: Accuracy and identity stability are separate properties

Both independent groups identify ‘Black’ and ‘Muslim’ as the most over-flagged categories, showing gender as a near-neutral factor.

Phase 1 findings

- 95% accuracy, but more false positives than false negatives
- High confidence doesn’t mean correct
- Long comments lost to 128 token cut
- Ethnicity and religion words alone push neutral text toward ‘Toxic’

Questions for Phase 2

How are the behavior patterns in phase 1 reflected in fine-tuned bert’s internal representations and token level sensitivity?

- Do identity terms sit near toxic comments in embedding space?
- Which tokens drive the flipped predictions?
- Does the shortcut appear in early or later layers?

Thank You!